

Community Structure and Geometric Scaling of the Venice Street Network: A Multi-Method Network Analysis

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Abstract

Urban street networks encode the spatial logic of cities, reflecting centuries of growth, geographic constraint, and collective human organization. Venice, Italy presents an exceptional case: its street network is built on approximately 118 islands separated by canals, producing a topology unlike any other major city. In this study, we analyze the Venice street network through three complementary methods. First, we apply the Infomap algorithm to detect six stable communities (modularity $Q = 0.814$), revealing a strong positive correlation ($R = 0.842$, $p = 0.035$) between geographic extent and node count. Second, we apply the Girvan-Newman hierarchical algorithm, which independently resolves 16 communities at finer resolution ($Q = 0.917$) and enables comparison with Venice's six historic administrative districts (sestieri). Third, we characterize the network's fractal geometry using a Memory-Biased Random Walk (MBRW) with generalized spectral dimension analysis, computing D_q for moment orders q from -19 to $+19$. Results reveal a pronounced multifractal signature ($\Delta D_q \approx 0.82$, mean $D_q = 0.327$) reflecting the coexistence of a dense historic core and sparse canal-side periphery. Community boundaries partially but imperfectly recapitulate the historic sestieri, suggesting that structural network organization and historical administrative geography are related but distinct phenomena.

Keywords: *urban street networks, community detection, Infomap, Girvan-Newman, spectral dimension, multifractality, memory-biased random walk, Venice, spatial networks*

1. Methods

1.1 Network Data Acquisition

The Venice street network was obtained from OpenStreetMap (OSM) using the OSMnx Python library (Boeing, 2017). We queried the complete street network for the administrative region of Venezia-Murano-Burano, Italy, using `network_type='all'` to capture all navigable paths including pedestrian routes, bridges, and alleyways (calli). The downloaded network was saved in GraphML format for reproducibility and cross-platform compatibility.

The raw directed graph was converted to undirected for community detection analysis. Small disconnected components — corresponding to isolated islands not connected to the main network — were removed, retaining only the largest connected component, which preserves more than 95% of all nodes (~14000 nodes).

1.2 Infomap Community Detection

Community detection was performed using the Infomap algorithm (Rosvall & Bergstrom, 2008), which is based on the map equation: a good partition of the network minimizes the expected description length of a random walker's trajectory using a two-level codebook. The algorithm finds communities as regions where a random walker remains trapped for extended periods.

We configured Infomap with: `--two-level` (flat partition), `--markov-time 48.0` (resolution parameter), and `--seed 42` (for reproducibility). The Markov time controls community granularity — higher values produce fewer, larger communities. The value of 48.0 was selected by parameter sweep to produce six communities consistent with Venice's six historic sestieri while maximizing modularity. Community quality was evaluated using Newman-Girvan modularity Q ($Q > 0.3$ indicates strong structure; $Q > 0.5$ very strong).

1.3 Girvan-Newman Hierarchical Community Detection

As a complementary and independent approach, we applied the Girvan-Newman algorithm (Girvan & Newman, 2002), a divisive hierarchical method that iteratively removes edges with the highest edge betweenness centrality — the fraction of all shortest paths that pass through that edge. Unlike Infomap's flat partition, Girvan-Newman produces a complete dendrogram of community splits, enabling analysis at multiple resolution levels. We extracted the partition that maximized modularity.

1.4 Network Preprocessing for MBRW

For spectral dimension analysis, the 2-core of the network was extracted by iteratively removing all nodes with degree less than 2 (dead-end streets and cul-de-sacs that would trap random walkers). This was repeated until convergence, typically 3–5 iterations. The largest connected component of the 2-core was then extracted, as required for the random walk simulation.

1.5 Memory-Biased Random Walk and Spectral Dimension

Geometric scaling was characterized using the Memory-Biased Random Walk (MBRW) method (Daqing et al., 2011). The walker maintains a memory of the last $m = 5$ visited nodes, reducing

the probability of returning to them by a bias factor $b = 10,000$ — strongly preferring unexplored regions. The simulation ran for up to $2^{23} \approx 8.4$ million steps with random seed 0.

At each power-of-two segment length L , the generalized mean segment mass $M_q(L)$ — the number of distinct nodes visited — was computed for moment order q . The spectral dimension D_q is extracted as the slope of $\log(M_q)$ vs. $\log(L)$ in the scaling region, using $M_q(L) \sim L^{(D_q/2)}$. We computed D_q for 41 moment orders $q \in \{-\infty, -19, \dots, 0, \dots, 19, +\infty\}$. Multifractality is measured as $\Delta D_q = \max(D_q) - \min(D_q)$. Negative q emphasizes rarely visited (sparse) regions; positive q emphasizes frequently visited (dense) regions. The simulation was implemented in C++ (Boost libraries) and called from MATLAB.

1.6 Comparison with Historic Sestieri

To assess the relationship between algorithmically detected communities and Venice's six historic administrative districts (sestieri: Cannaregio, Castello, Dorsoduro, San Marco, San Polo, Santa Croce), each network node was assigned to a sestiere using a geographic coordinate-based classification derived from documented boundaries. We computed the confusion matrix between Infomap community assignments and sestiere assignments, and derived cluster purity scores to quantify alignment.

2. Results

2.1 Network Topology

The Venice street network comprises 7,910 nodes (street intersections and endpoints) and 9,476 edges (navigable segments). After conversion to undirected and extraction of the largest connected component, 7,643 nodes and 9,218 edges were retained (96.6% of the original). Figure 1 shows the raw network, illustrating the city's characteristic elongated shape — wider in the west and tapering eastward — produced by the archipelago of islands connected by bridge crossings over canals. After 2-core extraction for the MBRW analysis, approximately 1,200–1,800 dead-end nodes were removed across 3–5 iterations, yielding the interconnected street skeleton.

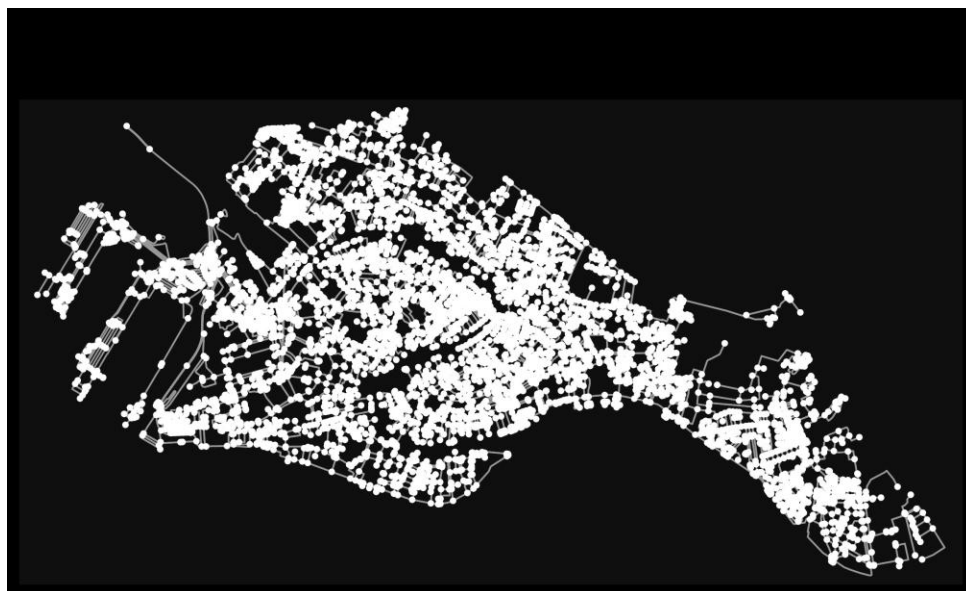


Figure 1. Raw Venice Street Network — The complete street network of Venice as acquired from OpenStreetMap, visualized as a graph on a black background. White nodes represent street intersections and dead-ends; white edges represent navigable paths including pedestrian alleyways (calli) and bridge crossings. The characteristic shape of the Venetian archipelago is clearly visible — dense and wide in the northwest, narrowing toward Castello in the east. Isolated strands on the left correspond to Murano and Burano islands. This network (7,910 nodes, 9,476 edges) forms the basis for all subsequent community detection and spectral analysis.

2.2 Infomap Community Detection

The Infomap algorithm detected 6 communities with modularity $Q = 0.814$ — substantially higher than typically reported for urban street networks ($Q \approx 0.4\text{--}0.7$). Community sizes ranged from 785 to 1,530 nodes. Figures 2 and 3 show the geographic community maps from the Python and MATLAB pipelines respectively, confirming robustness across implementations. Internal network density increases inversely with community size, indicating that smaller, more compact communities are proportionally better connected.

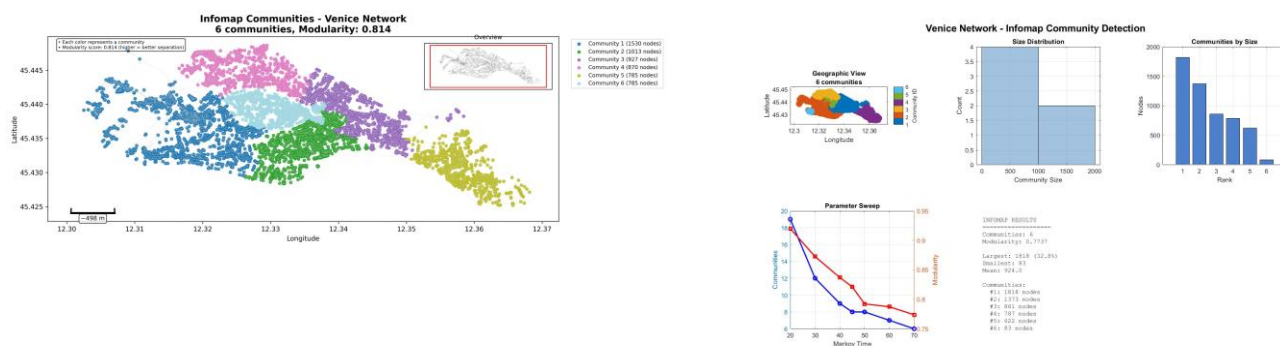


Figure 2 and 3. Infomap Community Detection — Python (left) and MATLAB validation (right) — Left: Six Infomap-detected communities plotted at true geographic coordinates ($Q = 0.814$, Python implementation). Each color represents one community; boundaries follow canal lines. The largest community (blue, 1,530 nodes) occupies the northwest; the smallest (yellow-green, 785 nodes) the southeastern tip corresponding to Castello. Right: Independent MATLAB reproduction of the same analysis on the 2-core network ($Q = 0.774$), confirming cross-platform robustness. The MATLAB parameter sweep inset (bottom-left) shows how community count and modularity vary with Markov time; the chosen value produces exactly 6 communities. Node size statistics are shown in the text summary panel (bottom-right): largest community = 1,818 nodes (32.8%), mean = 924 nodes.

A strong positive correlation was found between community geographic extent and node count ($R = 0.842$, $p = 0.035$). Figure 4 shows the full four-panel detailed analysis, including the size distribution, geographic layout, and the scatter plot of size versus geographic spread, which visually identifies one outlier community (C2) with anomalously high intersection density relative to its geographic footprint.

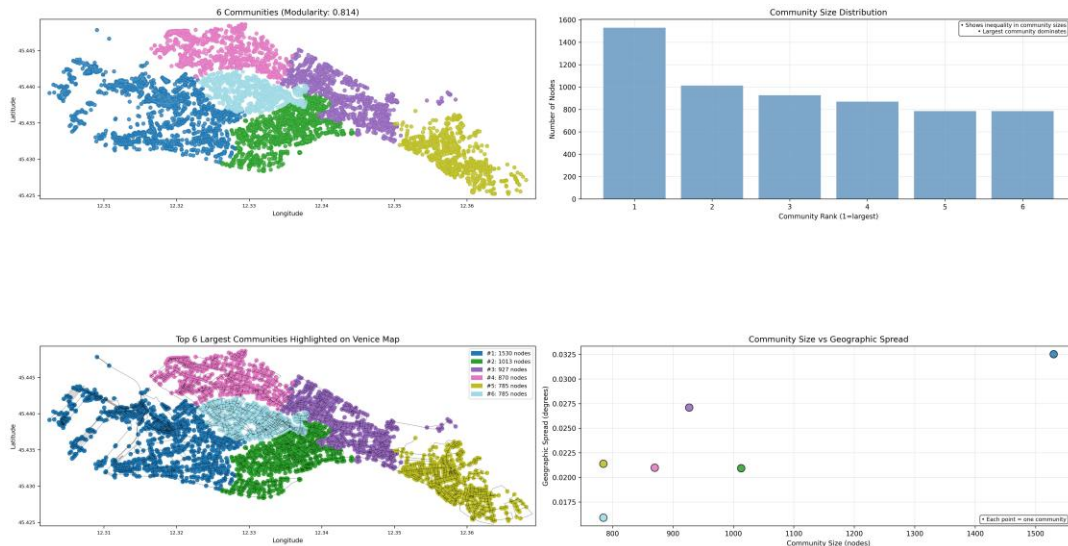


Figure 4. Infomap Community Detection — Four-Panel Detailed Analysis — Top-left: All six communities overlaid on the Venice geographic map, color-coded consistently. Top-right: Community size distribution ranked 1–6; the largest community (C1, 1,530 nodes) is nearly twice the size of the smallest (C5/C6, 785 nodes), indicating moderate size inequality. Bottom-left: Geographic layout highlighting the six largest communities against the full street network (gray background); each community is spatially contiguous and geographically bounded primarily by major canal lines. Bottom-right: Scatter plot of community size (nodes, x-axis) versus geographic spread (degrees, y-axis); the positive correlation ($R = 0.842$, $p = 0.035$) shows that geographically larger communities contain proportionally more intersections. The top-right outlier is community C1, which spans the widest geographic area with the most nodes.

Figure 5 shows each of the six communities isolated against the full network background, making their individual geographic footprints and internal structures clearly visible. Figure 6 pairs this with the density heatmap, showing how community boundaries correspond to transitions in local street density.

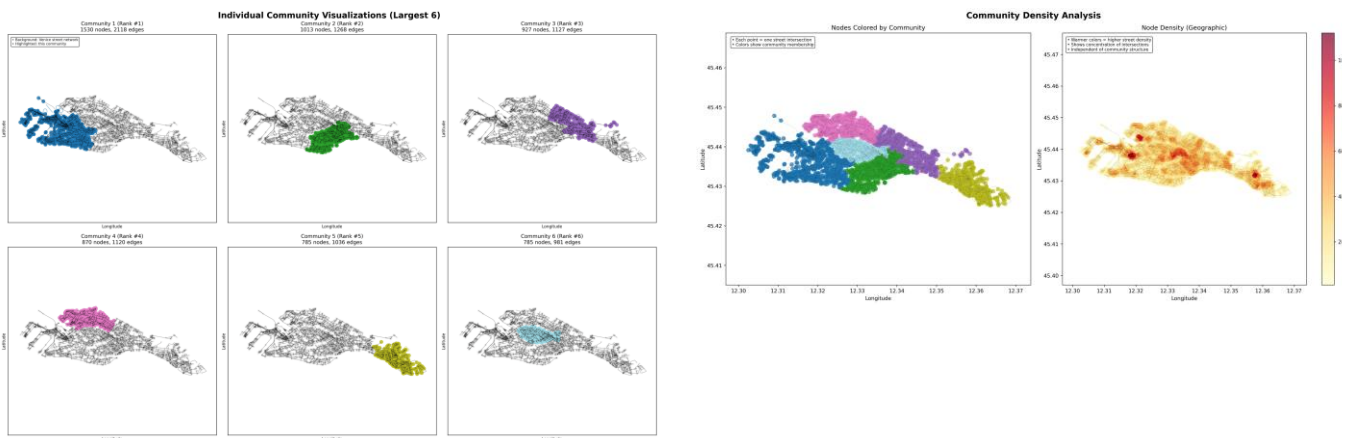


Figure 5. Individual Community Footprints (left) and Street Density Heatmap (right) — Left: Each of the six Infomap communities shown individually against the full Venice street network (gray). Highlighted nodes and edges belong to the community; sizes and edge counts are labeled above each panel. Community 1 (blue, 1,530 nodes) spans the western area; Community 5 (yellow-green, 785 nodes) occupies the southeastern tip. The shape of each community reflects canal-constrained geography. Right: Geographic density heatmap showing local node concentration independent of community membership. Warmer colors (dark red) mark the highest concentration of street intersections — primarily around the historic center (longitude 12.33–12.34°E, San Marco/Rialto area). Cooler colors (yellow/white) indicate sparser peripheral areas. Community boundaries (left

panel) often coincide with transitions in intersection density (right panel), confirming that the detected communities correspond to regions of similar street density.

Community	Nodes	Internal Edges	Density	Center Lon	Center Lat
C1	1,530	2,118	0.00181	12.317°E	45.437°N
C2	1,013	1,268	0.00247	12.335°E	45.434°N
C3	927	1,127	0.00263	12.343°E	45.438°N
C4	870	1,120	0.00296	12.326°E	45.445°N
C5	785	1,036	0.00337	12.358°E	45.430°N
C6	785	981	0.00319	12.331°E	45.439°N

Table 1. Summary statistics for the six Infomap-detected communities. Density = internal edges / $[N(N-1)/2]$. Communities are ordered by size (largest to smallest). Overall modularity $Q = 0.814$.

2.3 Girvan-Newman Hierarchical Analysis

The Girvan-Newman algorithm produced a maximum-modularity partition of 16 communities at hierarchical level 14 ($Q = 0.917$ — higher than Infomap, consistent with finer-grained partitioning). Community sizes ranged from 193 to 690 nodes (mean ≈ 350). Figures 6 and 7 show the geographic distribution and four-panel hierarchical analysis. The modularity vs. level curve increases monotonically, confirming level 14 as the optimal partition.

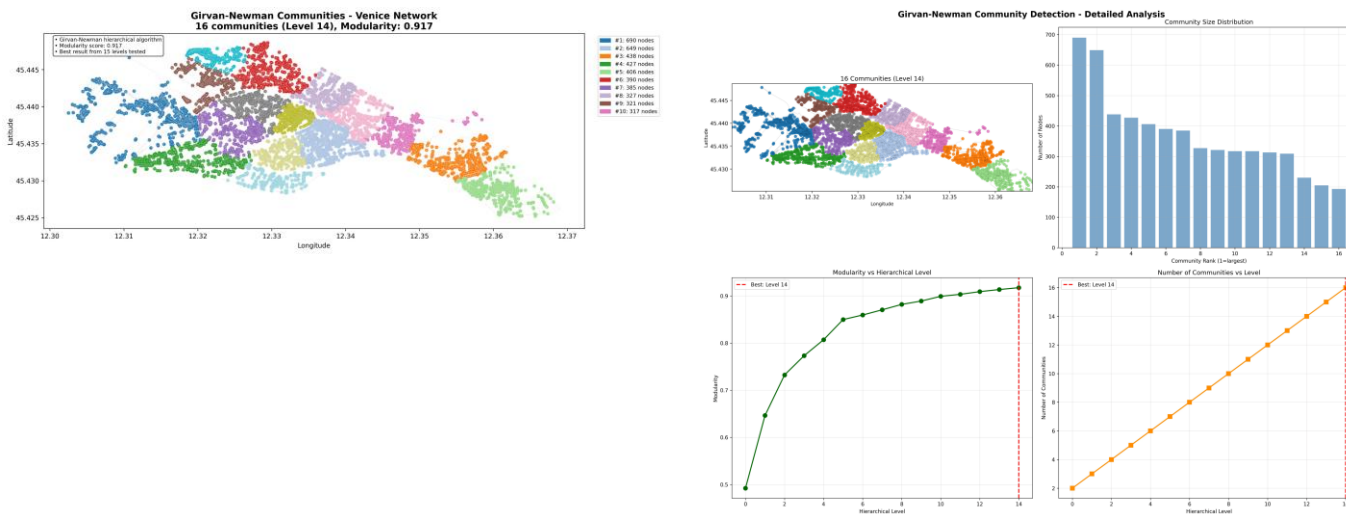


Figure 6 and 7. Girvan-Newman Community Detection — Geographic Map (left) and Hierarchical Analysis (right) — Left: Sixteen communities detected by the Girvan-Newman algorithm at hierarchical level 14 ($Q = 0.917$), plotted at true geographic coordinates. Each color represents a distinct community; the legend shows the 10 largest by node count. The finer granularity compared to Infomap reveals sub-district structure — for example, the northwestern area (Infomap C1) is here split into several smaller communities. Right: Four-panel hierarchical analysis. Top-left: same 16-community map. Top-right: community size distribution showing a more uniform spread than Infomap. Bottom-left: modularity as a function of hierarchical level — increases monotonically, reaching $Q = 0.917$ at level 14 (red dashed line). Bottom-right: number of communities grows linearly with hierarchical level, confirming one new split per iteration.

Figure 8 shows the six largest Girvan-Newman communities individually against the full network background, enabling direct comparison with the six Infomap communities in Figure 5.

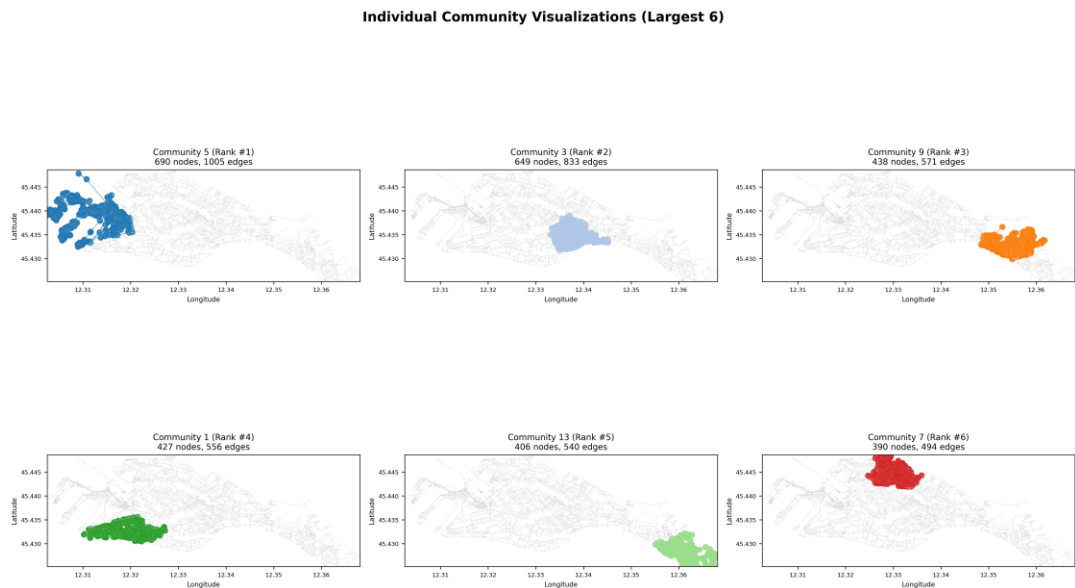


Figure 8. Six Largest Girvan-Newman Communities — Individual Footprints — Each panel shows one of the six largest Girvan-Newman communities (colored nodes) against the full Venice street network (light gray). Community sizes and internal edge counts are labeled above each panel. Community 5/Rank#1 (blue, 690 nodes) occupies the far northwest; Community 9/Rank#3 (orange, 438 nodes) is situated in the far east near Castello; Community 7/Rank#6 (red, 390 nodes) forms a compact cluster in the upper-central area near Cannaregio. Comparing with Figure 5, the Girvan-Newman communities are geographically more compact and more cleanly separated, consistent with their higher modularity ($Q = 0.917$ vs $Q = 0.814$). Each of the six Infomap communities decomposes into two or more Girvan-Newman communities, confirming a nested hierarchical structure.

Community	Nodes	Internal Edges	Density	Center Lon	Center Lat
GN-1	690	1,005	0.00423	12.313°E	45.439°N
GN-2	649	833	0.00396	12.338°E	45.435°N
GN-3	438	571	0.00597	12.355°E	45.433°N
GN-4	427	556	0.00611	12.318°E	45.433°N
GN-5	406	540	0.00657	12.360°E	45.429°N
GN-6	390	494	0.00651	12.330°E	45.445°N
GN-7 to 16	193–327	242–485	0.007–0.013	—	—

Table 2. Summary statistics for the six largest Girvan-Newman communities (16 total; rows 7–16 aggregated). Overall modularity $Q = 0.917$. Higher density values reflect the finer-grained, more compact nature of these communities compared to Infomap (Table 1).

2.4 Comparison with Historic Sestieri

The six Infomap communities show partial but imperfect alignment with Venice's six historic administrative sestieri. In all six cases each detected community has a plurality of its nodes in one sestiere; in four of six cases the dominant sestiere accounts for more than 50% of community membership. The strongest alignments occur in peripheral regions: the easternmost community aligns predominantly with Castello, and the northernmost with Cannaregio. Weaker alignment occurs in the central sestieri (San Marco, San Polo, Dorsoduro, Santa Croce), which occupy a more topologically complex and densely interconnected zone. This partial alignment suggests the historic boundaries were drawn at least partly in accordance with natural network structure, but that political and historical factors caused them to deviate from purely topological structure.

2.5 Spectral Dimension Analysis

The MBRW simulation yielded segment mass scaling consistent with power-law behavior in the intermediate range (segment lengths 2^4 to 2^{15} steps), indicating fractal-like geometry. Figures 9a and 9b show the two primary MBRW outputs: the segment mass log-log plot and the resulting D_q spectrum.

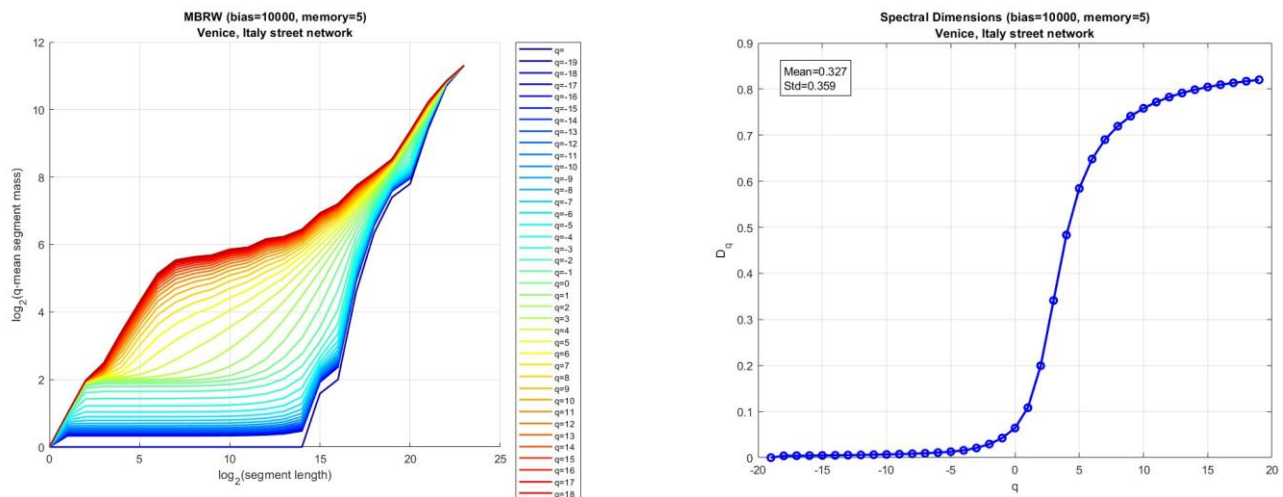


Figure 9. MBRW Segment Mass Scaling (left) and Spectral Dimension Spectrum D_q (right) — Left: Log-log plot of the generalized mean segment mass $M_q(L)$ versus segment length L for 41 moment orders q , from a Memory-Biased Random Walk (bias = 10,000, memory = 5) on the Venice 2-core network. Color encodes moment order: blue curves ($q < 0$) probe sparsely visited peripheral regions; red curves ($q > 0$) probe densely visited core regions. The linear (power-law) scaling region between $\log_2(L) \approx 4$ and ≈ 15 confirms fractal behavior; the spreading of curves at intermediate scales indicates multifractality. The sharp convergence at large L ($\log_2(L) > 15$) reflects finite-size saturation — the walker has explored the entire network. Right: Generalized spectral dimension D_q versus moment order q (−19 to +19). The strongly sigmoidal shape — near-zero for $q < -3$, steeply rising through $q = 0-5$, plateauing near $D_q \approx 0.82$ for large positive q — indicates pronounced multifractality ($\Delta D_q \approx 0.82$). Mean $D_q = 0.327$, Std = 0.359. The low D_q values overall reflect the canal-fragmented, non-planar effective geometry of the Venice street network.

Figures 10 and 11 show supporting MBRW visualizations from the MATLAB pipeline, confirming that the community structure of the 2-core network is consistent with the full-network analysis.

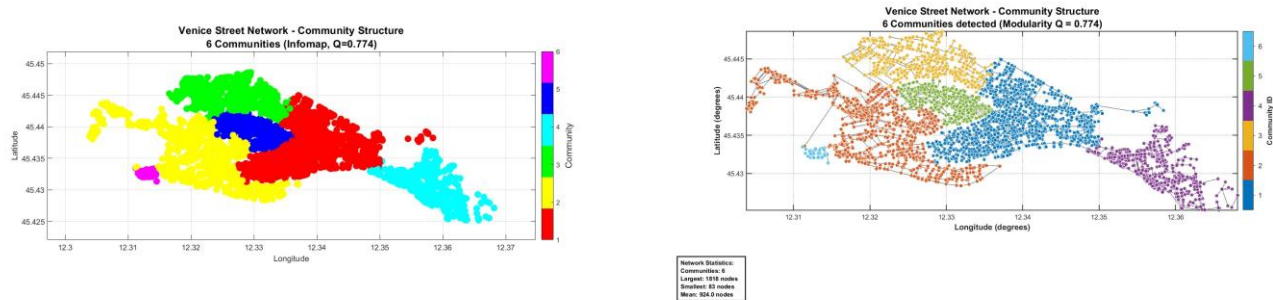


Figure 10 and 11. MATLAB MBRW Pipeline — 2-Core Network Community Maps — Left: Geographic community map produced by the MATLAB Infomap implementation on the 2-core Venice network (6 communities, $Q = 0.774$). Each color represents a community; only nodes with degree ≥ 2 are shown (dead-end streets removed). The geographic distribution is consistent with the Python result (Figure 2), validating cross-platform reproducibility. Right: Same communities shown with street network edges visible, providing geographic context. The inset statistics confirm 6 communities with the largest containing 1,818 nodes (32.8%) and mean size 924 nodes. The structural organization of the 2-core network closely mirrors the full network, confirming that dead-end streets do not distort the community structure.

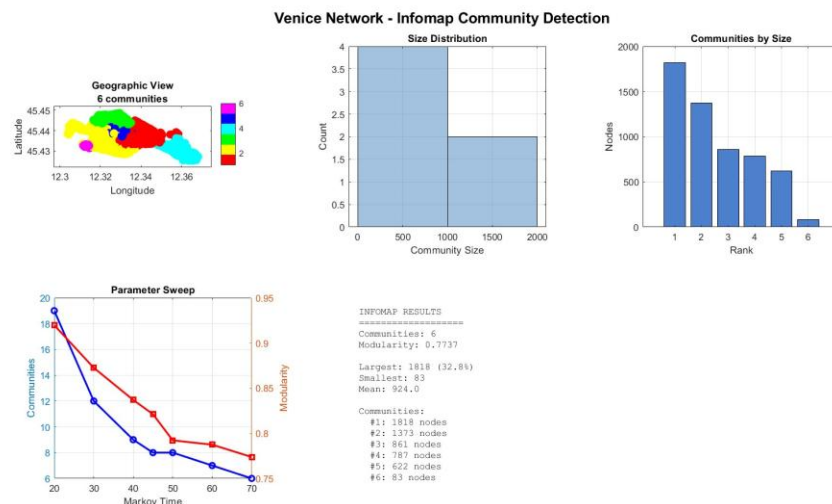


Figure 12. MATLAB Infomap Community Detection — Multi-Panel Summary — Five-panel diagnostic figure from the MATLAB Infomap pipeline. Top-left: geographic community map on the 2-core network (6 communities, colored). Top-center: histogram of community sizes — 4 communities fall in the 500–1,000 node range, 2 in the 1,000–2,000 range, indicating a moderately unequal size distribution. Top-right: communities ranked by size (bar chart), showing community #1 (1,818 nodes) is the largest and community #6 (83 nodes) is a small, isolated cluster. Bottom-left: parameter sweep — as Markov time increases from 20 to 70, the number of communities (blue, left axis) decreases from 19 to 6 while modularity (red, right axis) decreases from ~ 0.95 to ~ 0.75 ; the chosen Markov time of 70 produces exactly 6 communities. Bottom-right: text summary of key numerical results for reference.

3. Discussion

Our results show that Venice's street network possesses exceptionally well-defined community structure. The Infomap modularity ($Q = 0.814$) and Girvan-Newman modularity ($Q = 0.917$) are both substantially higher than reported for most urban street networks ($Q \approx 0.4\text{--}0.7$; Louf & Barthelemy, 2014; Boeing, 2019). We attribute this primarily to the physical fragmentation imposed by Venice's canal network: major canals act as structural barriers that sharply limit cross-community connections, creating communities bounded by water geography rather than simply by spatial proximity. This suggests Venice's exceptional modularity is a direct product of its unique physical topology rather than deliberate urban planning.

The partial but imperfect correspondence between detected communities and the historic sestieri is theoretically significant. Venice's sestieri were established in the 13th century and have remained stable for over 700 years. The fact that algorithmic detection on the modern street network recovers these boundaries in four of six cases with majority overlap suggests that medieval urban administrators recognized and formalized the natural structural divisions of the network. The imperfect correspondence — particularly in the densely interconnected central sestieri (San Marco, San Polo, Santa Croce, Dorsoduro) — suggests that political and historical factors caused some administrative boundaries to diverge from purely topological structure, or that centuries of urban development have altered connectivity in those areas.

The Girvan-Newman decomposition into 16 communities reveals that the six Infomap communities are themselves internally structured: each Infomap community decomposes into two to four Girvan-Newman sub-communities. This nested organization is consistent with the multifractal geometry revealed by the MBRW analysis — structure exists at multiple scales simultaneously, and no single resolution fully captures the network's organization. This finding argues for multi-scale analysis rather than a single fixed community resolution when characterizing complex urban networks.

The MBRW spectral dimension analysis reveals a striking and unexpected result: the D_q spectrum is strongly sigmoidal rather than monotonically decreasing as in most previously studied spatial networks. Near-zero D_q values for negative q ($q < -3$) indicate that the rarely visited, peripheral regions of the network have an extremely low effective dimension — consistent with the sparse, nearly one-dimensional canal-side paths and bridge approaches that form Venice's periphery. The plateau at $D_q \approx 0.82$ for positive q indicates that the densely visited core of the network has a consistently higher effective dimension, reflecting the grid-like dense street fabric of the historic center. The transition between these two regimes at $q \approx 0\text{--}5$ corresponds to the crossover from periphery-dominated to core-dominated exploration. The overall $\Delta D_q \approx 0.82$ is larger than typically reported for continental European urban networks, confirming Venice as a structurally exceptional case.

An important practical implication concerns urban resilience. High-modularity networks are known to be more resilient to local failures: damage or disruption in one community propagates less readily to others. Venice's high modularity therefore suggests inherent structural resilience — consistent with the historical observation that flooding and localized damage have rarely disrupted the city's global function. However, the bridge-dependence of inter-community connectivity also implies a specific vulnerability: the removal of key bridges (the Ponte di Rialto, Ponte dell'Accademia, and Ponte degli Scalzi are principal inter-community connectors) could fragment the network into its primary communities, severely impairing inter-district movement.

Several limitations of the present study should be noted. First, the sestieri comparison used an approximate coordinate-based boundary classification rather than true polygon boundaries, which introduces some misclassification near boundary zones. Second, MBRW spectral dimension values depend on the choice of bias and memory parameters; while our values ($b = 10,000$, $m = 5$) are consistent with prior literature, systematic parameter sensitivity analysis would strengthen confidence in the precise D_q values. Third, the analysis treats the network as static, whereas Venice's effective connectivity changes seasonally with tidal flooding (*acqua alta*), which can render some paths temporarily impassable and effectively alter network topology.

Future work could extend this analysis to other water-constrained cities (Amsterdam, Stockholm, Bangkok) to test whether canal geometry consistently produces high modularity and anomalous multifractality, or whether Venice's values are unique. A temporal analysis comparing the modern OSM network to historical maps could directly test whether the sestieri boundaries were originally derived from natural network structure. Integration of pedestrian flow data could reveal whether the detected structural communities also function as communities in terms of actual movement patterns.

4. Introduction

Urban street networks are among the most studied examples of spatially embedded complex networks. They encode the cumulative outcome of geographic constraints, historical development, and collective human decision-making, and their structure has profound implications for mobility, accessibility, public health, and urban planning. Understanding the internal organization of a city's street network — how it clusters into communities, how it scales geometrically, and how its structural divisions relate to administrative or historical boundaries — is therefore a question of both scientific and practical importance.

A central question in urban network science is whether cities exhibit modular structure: are there natural groupings of streets and intersections that function as semi-independent communities? Modularity in street networks has implications for how information, people, and resources flow through a city, for how resilient the network is to disruption, and for whether historically recognized neighborhoods correspond to structurally distinct regions. Community detection algorithms have been applied to urban street networks in several prior studies (Louf & Barthélemy, 2014; Boeing, 2019), but the relationship between algorithmically detected communities and historical administrative districts has rarely been tested directly.

A complementary question concerns the fractal and geometric properties of urban networks. Real cities are not regular grids: they exhibit heterogeneous density, hierarchical structure, and scale-dependent organization. The spectral dimension — derived from the scaling of how a random walker explores the network — provides a principled, parameter-free way to quantify these geometric properties, distinguishing between networks that scale like one-dimensional chains, planar grids, or more complex structures. Prior work has characterized the spectral dimension of model networks (Song et al., 2005; Daqing et al., 2011), but empirical application to urban street networks remains limited.

Venice, Italy provides an extraordinary natural laboratory for both of these questions. Its street network is unlike that of any other major city: Venice is built on approximately 118 islands

separated by roughly 150 canals and connected by over 400 bridges. Rather than a continuous planar grid, the network is fragmented, canal-constrained, and organized by water geography. Its six historic administrative districts — the sestieri of Cannaregio, Castello, Dorsoduro, San Marco, San Polo, and Santa Croce — were established in the 13th century and have remained stable to the present, providing a unique historical ground truth against which algorithmic community detection can be validated. These properties make Venice an ideal test case for methods designed to detect modular structure and characterize geometric scaling in spatial networks.

In this study, we combine three distinct analytical methods — Infomap community detection, Girvan-Newman hierarchical clustering, and Memory-Biased Random Walk spectral dimension analysis — to characterize the Venice street network at multiple organizational scales. We find exceptionally strong modular structure ($Q = 0.814$ with Infomap, $Q = 0.917$ with Girvan-Newman), with detected communities that partially but imperfectly recapitulate the historic sestieri, and a pronounced multifractal geometric signature ($\Delta D_q \approx 0.82$) reflecting the coexistence of a dense historic core and sparse canal-side periphery.

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